When the engine is off, the spring makes the plunger to sit on the seat. Once the engine is started, depending on the pressure and spring tension, spring will open or close the path by moving the plunger. When the engine speeds up, the oil pressure also increases and if the increasing pressure level reaches beyond the fixer level the spring presses the plunger from its seat so that the extra oil will reach the inlet via bypass or reaches the oil sump.

With the help of the adjusting shim and lock nut the tension of the spring can be adjusted. By adding or removing the shim the spring tension is increased or decreased

Oil Dipstick or Oil Level Indicator:

Oil dipstick is used to measure the oil level in the oil sump. A cap is placed at the top and the Steel rod or a blade is placed at the bottom. This rod will have two Limits as maximum and minimum level lines. Apart from this many small graduations are drawn below the minimum level line; oil should be maintained above minimum level line.

The dip stick is inserted through the engine block into the oil sump. Before starting the engine the dipstick has to be taken out and checked for the oil level and should be seen that the oil level should not go below the minimum level. Lubricating failures and reason for failures have to be taken in care for proper maintenance of the engine

8.7 FAILURES IN LUBRICATING SYSTEM

Following are the some of the failures in lubricating system

1. Low engine oil level
2. Interior cracks
3. Poor lubrication efficiency
4. Blockage of filters
5. Impurities in the Engine oil

8.8 REASONS FOR FAILURE IN LUBRICATING SYSTEM

There are many reasons for failures of lubricating system

1. Internal and external leakage of oil leads to reduced oil level
2. Low pressure in oil, the oil pump may get damaged
3. Failure of oil pump belt or chain
4. Blockage in filter or bend in pipe
5. More blocks in oil filter
6. Damaged compounds
7. Broken or burnt gasket
8. Broken or worn out Piston rings

8.9 METHODS TO TROUBLESHOOT

1. By correcting the oil leakage
2. By changing the gaskets in oil sump
3. By changing the head gasket
4. By changing the paper gasket cover
5. By changing the drain block
6. By changing the piston rings
7. By maintaining the correct oil level

Student Activity

1. Students should visit the nearby workshops to study the process of engine lubrication system.
2. Students should dismantle any on of the lubrication system and sketch the complete system and then describe their function in detail.

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